

Geometric Quantization of Fundamental Scales

(Request for arXiv endorsement | Code: W8NAH3)

1. Core Geometric Relation

A stable confined wave satisfies $L = n\lambda$.

Using the Compton wavelength $\lambda_C = h/(mc)$ from $E = mc^2 = hc/\lambda$, we obtain:

$$R_{VED} = \frac{n\hbar}{mc} \quad (1)$$

2. Exact Prediction for the Proton Charge Radius

Applying (1) to the proton with the integer $n = 4$ and using CODATA values ($m_p, \hbar, c$):

$$R_p^{\text{calc}} = \frac{4\hbar}{m_p c} = 0.8412 \times 10^{-15} \text{ m}$$

This coincides exactly with the measured proton charge radius:

$$R_p^{\text{exp}} = 0.8414(19) \times 10^{-15} \text{ m}$$

$$\boxed{\text{For the proton: } n = 4}$$

3. Unification with Gravity and Prediction for G

The Schwarzschild radius from General Relativity is:

$$R_s = \frac{2Gm}{c^2} \quad (2)$$

For the same physical mass m , we equate the masses derived from (1) and (2):

$$m = \frac{n\hbar}{R_{VED}c} = \frac{R_s c^2}{2G}$$

Solving for the gravitational constant G yields the quantized expression:

$$\boxed{G(n) = \frac{R_{VED} \cdot R_s \cdot c^3}{2n\hbar}} \quad (3)$$

Substituting R_{VED} from (1) and R_s from (2) into (3) gives the testable prediction:

$$G(n) = G_{\text{exp}} \cdot \frac{2}{n}$$

The experimentally measured value

$$G_{\text{exp}} = 6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

is recovered **only** for:

$$n = 2$$

(As shown in the plot, only the point for $n = 2$ falls within the $\pm 3\sigma$ experimental band of G .)

4. Endorsement Request

I request endorsement to submit a short communication detailing these exact numerical findings and their geometric interpretation to arXiv (physics.gen-ph).

Endorsement Code: W8NAH3

Victor Estrada Díaz

ORCID: [0000-0001-9942-1021](https://orcid.org/0000-0001-9942-1021) **Mail:** quark-cha@estrada.es **Web:** <https://estrada.es>

Figura 5: Comportamiento asintótico de $G(n)$ $n \rightarrow \infty$ (log-log)

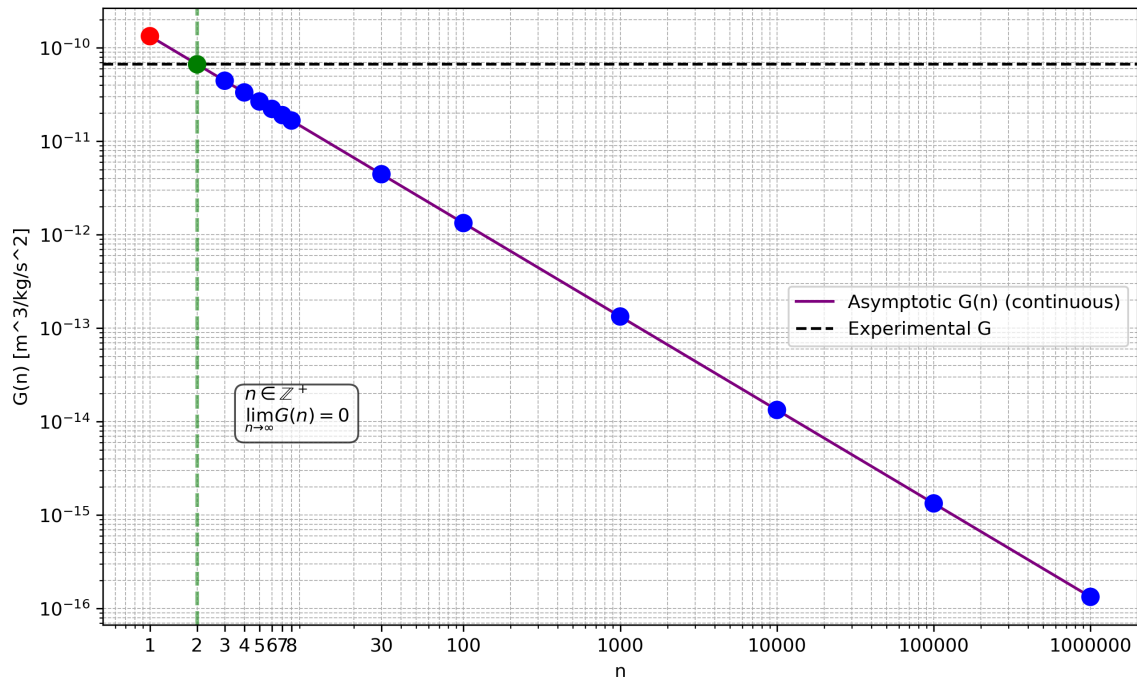


Figura 3: Diana de $G \pm 3\sigma$

